Company for now: Intel

Sector: Technology

Industry: Semiconductors

Frontend + Backend

1. Deal Intake + Document Upload

The analyst first enters the basic loan request:

- Borrower/company name
- Facility type
- Requested amount
- Purpose of loan
- Tenor
- Repayment structure
- Collateral/security, if applicable

Then they upload the relevant documents, such as annual reports, 10-Ks, 10-Qs, audited financial statements and other supporting files.

This is important because the AI should not simply analyse *"Company A"* generally. It should analyse whether **Company A can support this particular loan request**. BU5203 Group

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2. Extract and Organise the Relevant Information (Prompt Engineering)

The system reads all uploaded documents and extracts the information needed for the credit assessment.

Don't use an external database.

For example:

Business information

- What the company does
- Industry
- Business model
- Revenue sources
- Ownership/management
- Main customers or suppliers, where available

Financial information

- Revenue
- EBITDA
- Net income
- Cash
- Debt
- Receivables
- Inventory
- Operating cash flow
- Interest expense
- Capex

The extracted information should be converted into a **structured borrower dataset**, rather than repeatedly asking the AI to search through hundreds of PDF pages.

Most importantly, retain the **source and page number** for extracted information so that the analyst can verify it later. The second proposal makes this source-tracking part of the document extraction process.

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3. Calculate the Important Credit Metrics (Prompt Engineering)

Once the figures have been extracted, use normal programmed formulas to calculate the credit ratios.

For your prototype, I would **not calculate dozens of ratios**. Around **6–8 important metrics** should be enough:

1. Revenue growth
2. EBITDA margin
3. Debt / EBITDA
4. Interest coverage
5. Current ratio
6. DSCR
7. Operating cash flow
8. Working capital / DSO where relevant

One particularly important idea across the PDFs is that **the LLM should not perform the mathematics itself**.

Your backend should calculate:

Debt / EBITDA = Total Debt ÷ EBITDA

and then give the result to the AI.

The AI's role is to **interpret the number**, not calculate it. This reduces hallucinations and calculation errors. BU5203 Group

For example:

> FY2023 Debt/EBITDA = 2.1x
> FY2024 = 2.8x
> FY2025 = 3.7x

The AI can then explain:

> "Leverage has increased materially over the three-year period."

The deterministic credit-engine idea is also explicitly included in the second proposal.

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4. Analyse Historical Trends and Repayment Capacity (Prompt Engineering)

The objective of 4. is to move beyond isolated ratio reporting to **trend-aware, relationship-based interpretation** of financials. Basically, Instead of looking at numbers individually, the AI looks for **relationships and trends**. The AI should detect *divergences* (e.g., revenue up but cash flow down) and translate them into **credit-relevant insights** about repayment capacity and risk.

Step 1: Two layers of Input fed to the AI, **(1) Extracted Business Information** and (2) Calculated Ratios + Time-Series

A. Extracted Business Information

- Business model and revenue drivers (who pays, why, how recurring vs one-off)
- Industry, geography, customer concentration, seasonality
- Capital intensity and working-capital dynamics
- Management quality and governance highlights (if available)

B. Calculated Ratios + Time-Series

For at least 3–5 years (or quarters), provide:

- **Profitability**: Revenue, EBITDA, EBITDA margin, net profit margin
- **Leverage**: Total debt, net debt, Debt/EBITDA, Debt/Equity
- **Liquidity**: Current ratio, quick ratio, cash balance
- **Cash flow**: Operating cash flow, capex, free cash flow
- **Debt service**: Interest expense, principal repayments, DSCR, interest cover

Structure this as a clean table or JSON so the AI can see **direction and magnitude of change** over time.

Step 2: Crafting how the AI will **(1) Analyse and Detect Trend Patterns and Divergences** and **(2) Link Trends to Business Context**

A. Detect Trend Patterns and Divergences

Ask the AI to classify each metric's trend (↑, ↓, ~) and then **look for conflicting signals**, e.g.:

- Revenue ↑, EBITDA ↓ → margin compression, cost pressure, pricing power issues
- Revenue ↑, Operating cash flow ↓ → working-capital strain, receivables build-up
- EBITDA ~, Debt ↑, DSCR ↓ → leverage rising faster than earnings, weakening debt-servicing capacity
- Liquidity ratios ↓ while short-term debt ↑ → refinancing and rollover risk

This is where your example fits neatly as a pattern the AI must recognise:

Revenue ↑
EBITDA ↓
Debt ↑
DSCR ↓

Interpretation: Top-line growth is not translating into profitability; leverage is increasing and debt-servicing capacity is deteriorating. Credit risk is rising despite growth.

B. Link Trends to Business Context

Require the AI to connect numerical trends to the business story:

- Is margin compression due to input costs, competition, or discounting?
- Is debt increasing for growth capex (potentially justified) or to fund operating losses (red flag)?
- Are liquidity pressures seasonal or structural?

This prevents purely mechanical ratio commentary and pushes toward **credit judgement**.

Step 3: Three-Question Framework for the AI Output

The AI Analysis should be structured around these 3 questions, **(1) Business Overview**, **(2) Financial Analysis** and **(3) Credit Risk**. This aligns with standard credit frameworks (5 Cs / CAMPARI-style logic) and keeps outputs decision-useful.

A. Business Overview — "Who are they?" - What does the company do and how does it generate money?

Prompt the AI to answer:

- What does the company do, and how does it generate money?
- What are the key revenue drivers and cost structure?
- How cyclical, competitive, or capital-intensive is the business?
- Any notable customer/supplier concentration or regulatory dependencies?

Output: 3–5 concise bullets that set the **context for the numbers**.

B. Financial Analysis — "Can they repay?" - Analyse the company's profitability, leverage, liquidity, cash flow and debt-servicing ability

Here, the AI uses the trend-aware logic:

- **Profitability**: Are margins stable, expanding, or compressing? How does EBITDA trend vs revenue?
- **Leverage**: Is Debt/EBITDA improving or deteriorating? Is the capital structure getting riskier?
- **Liquidity**: Can the firm meet short-term obligations without stress?
- **Cash flow & debt service**:
    - Is operating cash flow sufficient to cover capex and debt service?
    - What is the DSCR trajectory?
    - Are there signs of reliance on refinancing or asset sales?

Crucially, ask the AI to **state the narrative**, not just list ratios. For example:

"Although revenue has grown at 8% CAGR, EBITDA margin has contracted from 18% to 12%, indicating rising cost pressure. Coupled with a 40% increase in debt, DSCR has fallen from 1.6x to 1.1x, signalling materially weaker debt-servicing capacity."

C. Credit Risk — "What could stop them from repaying?" - Identify the main financial, business and external risks, matching the original simpler three-part backend concept.

Prompt the AI to identify and prioritise risks in three buckets:

1. **Financial risks**
    - High or rising leverage, weak DSCR, thin liquidity
    - Dependence on refinancing or covenant headroom erosion
2. **Business risks**
    - Customer concentration, margin pressure, competitive threats
    - Execution risk on growth plans or capex projects
3. **External risks**

    - Industry cyclicality, regulatory changes, macro shocks (rates, FX, commodity prices)
    - For Singapore or regional names: sector-specific risks (e.g., travel, property, trade)

Ask the AI to **rank** the top 2–3 risks and briefly explain why they matter for repayment.

Example Prompt Template (Steps 1 - 3)

You are a credit analyst AI. You will receive:

- (1) a short business overview, and
  (2) a time-series of key financial metrics and ratios (at least 3 periods).

Your task (basically the prompts):

- First, describe the business model and how it generates cash.
- Second, analyse profitability, leverage, liquidity, cash flow and debt-servicing capacity, focusing on **trends and relationships** between metrics (e.g., revenue vs EBITDA, debt vs DSCR). Highlight any divergences and what they imply for repayment capacity.
- Third, identify the main financial, business and external risks that could impair repayment. Rank the top 2–3 risks and justify your ranking.

Keep your output structured under three headings:

1. Business Overview – “Who are they?”
2. Financial Analysis – “Can they repay?”
3. Credit Risk – “What could stop them from repaying?”

Use clear, concise, presentation-ready language suitable for an investment or credit committee.

You can then plug in your extracted data and ratios as the user message.

How this improves Credit Analaysis vs Simple Ratio Reporting

- **From static snapshots to dynamic stories**: Instead of “Revenue increased 10%,” you get “Revenue increased 10%, but EBITDA fell 5%, implying margin compression and weaker cash generation.”
- **From isolated metrics to connected signals**: The AI explicitly links revenue, profitability, leverage, and DSCR to assess **repayment capacity**, not just performance
- **From description to decision support**: By forcing a risk section and ranking, the output becomes directly usable for credit decisions, covenant design, and monitoring.

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5. Identify Key Risks + Suggest Mitigants (Prompt Engineering)

The objective of 5. is to move from generic "risk lists" to a **focused, evidence-backed risk → mitigant → term** output. The AI should surface only the **most material 3–5 risks**, tie each to **specific data evidence**, and propose **practical mitigants and loan-term adjustments** for the analyst for CONSIDERATION ONLY, not for making automatic decisions!

Step 1: What the AI should produce (Output Structure)

Instruct the AI to output a compact table (or structured bullets) with four columns/fields per risk:

1. **Identified Risk** – concise label (e.g., "High leverage", "Weak liquidity").
2. **Evidence** – specific metrics/trends that justify the risk (with numbers and direction).
3. **Possible Mitigant** – actions that reduce probability or impact of the risk.
4. **Proposed Term / Monitoring Condition** – how the mitigant translates into loan structure, covenants, or reporting.

Example layout:

Identified Risk	Evidence	Possible Mitigant	Proposed Term / Condition
High leverage	Debt/EBITDA increased from 2.1x to 3.7x (3 years)	Limit additional debt; strengthen equity base	Maximum net leverage covenant (e.g., ≤3.5x); equity cure clause
Weak liquidity	Current ratio fell from 1.3x to 0.9x; cash/ST debt ↓	Ensure minimum cash buffer	Minimum liquidity covenant; monthly liquidity reporting
Weak cash conversion	Receivables days up from 45 to 78; outpacing sales	Tighten working-capital management	Maximum DSO covenant; receivables aging report monthly
Customer concentration	Top customer = 40% of revenue	Diversify customer base; monitor key account	Concentration reporting; notification if top customer >35%

Refinancing risk	60% of debt matures within 12 months	Stagger maturities; secure refinancing early	Shorter tenor / refinancing condition; mandatory hedging or amortisation

This mirrors your “Risk → Mitigant → Proposed Term” idea and keeps outputs decision-useful rather than descriptive e.g. *"high leverage is a risk."*.

Step 2: How the AI should select risks (Prompt Logic)

In order to avoid long, generic lists, engineer the prompt to **(1) Prioritise Materiality** and **(2) Tie each Risk to Specific Evidence**

A. Prioritise Materiality

Ask the AI to:

- Rank risks by **impact on repayment capacity** and **likelihood** (high/medium/low).
- Select only the **top 3–5 risks** that are both material and supported by clear evidence.

You can explicitly say:

“Identify up to five material credit risks that could impair repayment. Prioritise risks where there is clear quantitative evidence (ratio deterioration, trend divergence, concentration, maturity profile).”

B. Tie Each Risk to Specific Evidence

Require each risk to reference:

- Specific ratios or metrics (e.g., “Debt/EBITDA from 2.1x to 3.7x”).
- Time horizon (e.g., “over the last three fiscal years”).
- Any relevant business context (e.g., “in a cyclical industry with falling margins”).

This prevents vague statements like “high leverage is a risk” without showing *why* it matters here. Do remember that these should remain **suggestions for the analyst**, not automatic lending decisions.

Step 3: Designing Mitigants and Terms (Suggestions made by the AI)

For each identified risk, the AI should propose **(1) Mitigants/Risk-Reduction Actions**, **(2) Proposed Terms and Conditions aka Loan Structure Translation**

A. Mitigants (Risk-Reduction Actions)

Examples by risk type:

- **High leverage**
  - Limit further debt; require equity injection; restrict distributions.
- **Weak liquidity**
  - Maintain minimum cash; restrict certain outflows; improve cash forecasting.
- **Weak cash conversion / working capital stress**
  - Tighten credit terms; improve collections; monitor inventory and receivables.
- **Customer concentration**
  - Diversify customer base; deepen contracts with key customers; monitor churn.
- **Refinancing risk**
  - Stagger debt maturities; secure committed facilities earlier; consider amortising structures.

B. Proposed Terms / Conditions (Loan-Structure Translation)

Map each mitigant into something concrete a credit committee can use:

- **Covenants**
  - Maximum leverage ratio (e.g., Net Debt/EBITDA ≤ 3.5x).
  - Minimum interest cover or FCCR (e.g., FCCR ≥ 1.2x).
  - Minimum liquidity (e.g., cash + undrawn RCF ≥ X months of opex).
- **Reporting / Monitoring**
  - Monthly receivables aging and DSO report.
  - Quarterly customer concentration report (top 5 customers as % of revenue).
  - Refinancing status update when >25% of debt is due within 12 months.
- **Structural Protections**
  - Shorter tenor or mandatory amortisation.
  - Equity cure rights (limited frequency, cash-only).
  - Restrictions on distributions or related-party outflows as leverage rises.

Crucially, frame these as **suggestions for the analyst**, not automatic lending decisions. The AI's role is to surface options, not to approve or decline.

Example Prompt

You can implement something like this as your system or user prompt:\

You are a credit analyst AI. You will receive:

- (1) a short business overview, and
  (2) a time-series of key financial metrics and ratios.

  Your task:

- Identify up to **five material credit risks** that could impair the borrower's ability to repay. Prioritise risks with clear quantitative evidence (ratio trends, concentration, maturity profile).
- For each risk, provide:
    1. **Identified Risk** – a concise label.
    2. **Evidence** – specific metrics/trends and time period that support the risk.
    3. **Possible Mitigant** – practical actions that could reduce the probability or impact of this risk.
    4. **Proposed Term / Condition** – how the mitigant could be reflected in loan structure, covenants, reporting, or monitoring.

Keep your output structured as a compact table or clearly labelled bullets under each risk. These are **suggestions for the analyst**, not automatic lending decisions. Focus on clarity, relevance, and credit impact.

You then plug in your extracted business info and ratio trends as the user message.

5) How This Improves on Generic Risk Lists

- **From generic to specific**: Instead of "high leverage is a risk," you get "Debt/EBITDA rose from 2.1x to 3.7x; consider a max net leverage covenant and equity cure."
- **From description to action**: Each risk is paired with a mitigant and a concrete term/condition the deal team can negotiate.
- **From exhaustive to focused**: By limiting to 3–5 material risks, the output stays readable and decision-oriented for credit committees.

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6. Generate the Credit Memo With Evidence (Prompt Engineering)

Only after Steps 1–5 should the AI write the memo.

I would keep the memo structure relatively simple:

1. Executive Summary

- Borrower
- Facility requested
- Purpose
- Key strengths
- Key risks

2. Business Overview

- Business model

- Industry
- Revenue sources
- Management/ownership where relevant

3. Financial Analysis

- Revenue and profitability
- Cash flow
- Liquidity
- Leverage
- Historical trends

4. Repayment Capacity

- Cash generation
- DSCR
- Debt obligations

5. Key Credit Risks

- Financial
- Business
- Industry/external

6. Risk Mitigants / Proposed Structure

- Covenants
- Collateral
- Tenor
- Repayment structure
- Monitoring requirements

7. Information Gaps

- Missing documents
- Conflicting information
- Unsupported claims

Propose structure for the generated memo.

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7. Analyst Review + Source Verification (Important)

This should be one of the **main points**, we call this an “audit trail”.

Every important fact generated by the AI should link back to its original source to address hallucinations, explainability and analyst credit-risk.

For example:

> Revenue increased 10.5% in FY2025.
> **Source: FY2025 Annual Report, p. 23**

or:

> Debt/EBITDA = 3.7x.
> **Calculated using Total Debt from FY2025 Annual Report p. 65 and EBITDA from p. 42.**

This creates:

Source → Extracted Data → Calculation → AI Interpretation → Credit Memo

Analysts can therefore check where statements came from instead of blindly trusting the AI. The proposals explicitly emphasise this audit trail and human verification.

The claim-level citation idea in the larger proposal similarly argues that important memo statements should point directly to their supporting documents.

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1. Deal Intake + Upload Documents
↓
2. Extract Business & Financial Information
↓
3. Calculate Credit Ratios Using Code
↓
4. AI Analyses Business, Financial Trends & Repayment Ability
↓
5. AI Identifies Key Risks + Possible Mitigants
↓
6. AI Generates Structured Credit Memo
↓
7. Analyst Verifies Sources, Edits and Approves

Deliverables:

1. Engineering and Designing Prompt for 2-6
2. Frontend (GUI), Backend
3. Prototype Testing